

Note 3.3 Differentiation Rules

- **Constant Rule:**

$$(c)' = 0$$

- **Power Rule:**

$$(x^\alpha)' = \alpha x^{a-1}$$

- **Trigonometric:**

$$(\sin x)' = \cos x, (\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x, (\csc x)' = -\csc x \cdot \cot x$$

$$(\sec x)' = \sec x \cdot \tan x, (\cot x)' = -\csc^2 x$$

- **Logarithmic:**

- $(\log_a x)' = \frac{1}{x \ln a}$

- $(\ln x)' = \frac{1}{x}$

- **Exponential:**

- $(a^x)' = a^x \ln a$

- $(e^x)' = e^x$

- **The Constant Multiple Rule:**

If c is a constant and $f(x)$ is differentiable, then

$$\frac{d}{dx}(cf(x)) = c \frac{d}{dx} f(x)$$

- **The Sum Rule:**

If $f(x)$ and $g(x)$ are differentiable, then

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

The rule can be extended to the sum of finite terms of functions

- **The Product Rule:**

If $f(x)$ and $g(x)$ are differentiable, then

$$\frac{d}{dx}(f(x)g(x)) = f(x) \frac{d}{dx} g(x) + g(x) \frac{d}{dx} f(x)$$

or equivalently,

$$(fg)' = fg' + f'g$$

Corollary:

$$(uvw)' = u'vw + uv'w + uvw'$$

- **The Quotient Rule:**

If $f(x)$ and $g(x)$ are differentiable, then

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{g^2(x)} \text{ if } g(x) \neq 0$$

- **Higher-order Derivatives:**

- **The Second Derivatives:**

The second derivative of a function is the derivative of its derivative, if $y = f(x)$, it is denoted by

$$\frac{d^2y}{dx^2} \text{ or } f''(x)$$

- **The n th Derivatives:**

If the original function is $y = f(x)$, the n th derivative is denoted by

$$\frac{d^ny}{dx^n} \text{ or } f^{(n)}(x)$$